

## Unit 1 · Coordinates, vectors and components

Reference sheet · Positions, arithmetic and length

Edition 1 · 3 October 2026 · Page 1 of 2

### Coordinates and displacement

Choose an origin, positive axis directions and units. A point  $P = (x, y, z)$  has position vector  $\mathbf{r}_P = \langle x, y, z \rangle$ .

$$\vec{AB} = \mathbf{r}_B - \mathbf{r}_A.$$

Final minus initial, component by component. A fixed origin shift cancels if axes and units are preserved.

Endpoint data do not determine the route.

L01 §§1.1–1.6

### Vector arithmetic

$$\mathbf{a} \pm \mathbf{b} = \langle a_x \pm b_x, a_y \pm b_y \rangle.$$

$$k\mathbf{a} = \langle ka_x, ka_y \rangle.$$

Include the third component in 3D. Use a common frame and compatible units. Subtraction adds the opposite; a minus sign changes every component. Head-to-tail arrows show addition.

$$k(\mathbf{a} + \mathbf{b}) = k\mathbf{a} + k\mathbf{b}.$$

L02 §§1.8–1.13

### Magnitude and travelled distance

$$|\mathbf{v}| = \sqrt{v_x^2 + v_y^2} \quad (2D),$$

$$|\mathbf{v}| = \sqrt{v_x^2 + v_y^2 + v_z^2} \quad (3D).$$

Axes are mutually perpendicular; components use consistent units. Magnitude is nonnegative and has the vector's units. Add straight-stage lengths for a route's distance; take the magnitude of their vector sum for its endpoint separation.

L03 §§1.15–1.17

### Scaling and direction

$$|k\mathbf{v}| = |k| |\mathbf{v}|.$$

For nonzero  $\mathbf{v}$  and nonzero  $k$ : positive  $k$  preserves direction; negative  $k$  reverses it. Zero  $k$  gives the zero vector. Moving an arrow parallel to itself does not scale it.

L02 §1.12; L03 §1.17

### Unit vectors

$$\mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|}, \quad \mathbf{v} \neq \mathbf{0}.$$

$$|\mathbf{u}| = 1, \quad \mathbf{d} = L\mathbf{u} \quad (L > 0).$$

$\mathbf{u}$  is dimensionless and points along  $\mathbf{v}$ . Multiply by a desired length  $L$  to construct  $\mathbf{d}$ ; use  $-\mathbf{d}$  for the opposite direction. The zero vector has no direction and cannot be normalised.

L03 §§1.18–1.20

## Unit 1 · Coordinates, vectors and components

Reference sheet · Direction, angles and checks

Edition 1 · 3 October 2026 · Page 2 of 2

### Coordinate unit vectors

$$\begin{aligned}\mathbf{i} &= \langle 1, 0, 0 \rangle, & \mathbf{j} &= \langle 0, 1, 0 \rangle, \\ \mathbf{k} &= \langle 0, 0, 1 \rangle. \\ \mathbf{v} &= v_x \mathbf{i} + v_y \mathbf{j} + v_z \mathbf{k}.\end{aligned}$$

These basis vectors are dimensionless. Physical units belong to the coefficients. In 2D omit the third term and use two-component  $\mathbf{i}$  and  $\mathbf{j}$ .

L03 §1.19

### Standard angles and components

Standard angle  $\theta$  starts at positive  $x$  and turns anticlockwise. For a plane vector of magnitude  $L$ :

$$v_x = L \cos \theta, \quad v_y = L \sin \theta.$$

Direction unit vector:  $\langle \cos \theta, \sin \theta \rangle$ . Check the component signs against the quadrant. A horizontal angle describes only the horizontal part of a 3D vector; use its horizontal length in these formulas.

L04 §§1.23–1.24, 1.27

### Recovering a direction

Find the quadrant from the signs first. When both components are nonzero, find the acute reference angle

$$\alpha = \arctan\left(\frac{|v_y|}{|v_x|}\right).$$

In quadrants I, II, III, IV the standard angle is respectively  $\alpha$ ,  $180^\circ - \alpha$ ,  $180^\circ + \alpha$ ,  $360^\circ - \alpha$ . On the axes:  $+x \rightarrow 0^\circ$ ,  $+y \rightarrow 90^\circ$ ,  $-x \rightarrow 180^\circ$ ,  $-y \rightarrow 270^\circ$ . For zero, no direction is defined.

L04 §1.25

### Bearings and angle units

With  $x$  east and  $y$  north, bearing  $\beta$  turns clockwise from north:

$$\begin{aligned}v_x &= L \sin \beta, & v_y &= L \cos \beta. \\ \theta &= 90^\circ - \beta.\end{aligned}$$

Add or subtract whole turns as needed. One turn =  $360^\circ = 2\pi$  radians. Degrees  $\rightarrow$  radians: multiply by  $\pi / 180$ ; radians  $\rightarrow$  degrees: multiply by  $180 / \pi$ . Match calculator mode to angle units. An arc of length  $s$  on radius  $r$  subtends  $\theta = s / r$  radians.

L04 §§1.22, 1.26

### A short checking routine

Name the required quantity. Fix the axes and units. Sketch signs and direction before calculating. Use components for a sum, then reconstruct length or angle if requested. Keep extra digits until the end.

Check a displacement by adding it to the initial position; check components by recovering the supplied magnitude; check a unit vector by squaring and adding its components. A successful check supports the argument but does not replace it.

L01 §1.7; L03 §1.21; L04 §1.27